

JAVA ROAD MAP(A-Z)

Programming Foundations-

1. What is programming?
2. How computers work
3. Binary basics
4. Variables and memory
5. Algorithms
6. Flowcharts
7. Pseudocode
8. IDE setup
9. Command line basics

Tools-

1. Git
2. GitHub
3. IntelliJ IDEA
4. JDK installation

Core Java Basics-

1. Java Introduction
2. History of Java
3. Features of Java
4. JVM
5. JRE
6. JDK
7. Bytecode
8. Platform independence
9. Compilation process

First Java Program-

- main() method
- Compilation & execution
- Naming conventions
- Comments

Data Types-

1. Primitive types
2. Non-primitive types
3. Literals
4. Constants (final)
5. Type casting

Operators-

1. Arithmetic
2. Relational
3. Logical
4. Unary
5. Assignment
6. Bitwise
7. Ternary
8. Operator precedence

Input/Output-

1. Scanner
2. BufferedReader
3. Console formatting

Control Flow-

1. if
2. if-else
3. switch
4. loops
 - for
 - while
 - do-while
5. break/continue

Methods-

1. Method declaration
2. Parameters
3. Return types
4. Overloading
5. Varargs
6. Recursion
7. Call stack
8. Pass-by-value
9. Stack memory behavior

Arrays-

1. 1D arrays
2. 2D arrays
3. Jagged arrays
4. Searching
5. Sorting
6. Algorithms

7. Linear search
8. Binary search
9. Bubble sort
10. Selection sort
11. Insertion sort

Strings-

1. Learn
2. String pool
3. Immutable strings
4. String methods
5. Classes
6. String
7. StringBuilder
8. StringBuffer
9. Regex

Object Oriented Programming(OOP)-

1.Classes & Objects

- Constructors
- this
- Object references

2.Encapsulation

- Access modifiers
- Getters/setters

3.Inheritance

- extends
- super
- Constructor chaining

4.Polymorphism

- Compile-time
- Runtime
- Dynamic dispatch

5.Abstraction

- Abstract classes

- Interfaces

6.Relationships

- Association
- Aggregation
- Composition

Advance Core Java-

1.Packages

- Creating/importing packages
- Static imports

2.Exception Handling

- try-catch
- finally
- throw/throws
- Custom exceptions
- Checked vs unchecked exceptions

3.Inner Classes

- Static nested
- Anonymous
- Local inner classes

4.Enums

- Enum methods
- Enum constructors

Java Memory Management-

1. Stack memory
2. Heap memory
3. Method area
4. Garbage collection
5. Memory leaks

Collection Frameworks-

1.List

- ArrayList
- LinkedList
- Vector
- Stack

2.Set

- HashSet
- LinkedHashSet
- TreeSet

3.Queue

- PriorityQueue
- Deque

4.Map

- HashMap
- LinkedHashMap
- TreeMap
- Hashtable

5.Iterators

6.Comparable

7.Comparator

8.Generics

9.Autoboxing

10.Immutable collections

Functional Programming-

1. Lambda expressions
2. Functional interfaces
3. Method references

Streams API-

1. map
2. filter
3. reduce

4. collect
5. parallel streams
6. Optional class

Date & Time API-

1. LocalDate
2. LocalTime
3. LocalDateTime

Java I/O-

1. Byte streams
2. Character streams
3. Buffered streams

File Handling-

1. File class
2. Reading/writing files

Serialization-

1. Serializable
2. Object streams

NIO-

1. Path
2. Files
3. Buffers
4. Channels

Multithreading & Concurrency-

1. Basics

- Threads
- Runnable
- Thread lifecycle

2. Synchronization

- synchronized
- Race conditions
- Locks

3.Advanced

- Executor framework
- Thread pools
- Callable/Future
- Concurrent collections
- Atomic classes
- Volatile keyword

4.Modern

- CompletableFuture

JVM Internals-

1.JVM Basics

- JVM architecture
- Bytecode
- Class loading

2.Runtime Memory Areas

- Heap
- Stack
- Method Area
- PC Register
- Native Method Stack

4.Class Loaders

- Bootstrap
- Extension
- Application

5.Execution Engine

- Interpreter
- JIT Compiler
- 6.Garbage Collection
- G1GC
- ZGC
- CMS

- Generational GC

7.Java Memory Model (JMM)

- Happens-before
- Visibility
- Reordering
- Volatile

8.Performance Tuning

- Heap tuning
- GC tuning
- JVM flags

9.Monitoring Tools

- jvisualvm
- jconsole
- jstack
- jmap
- jcmd

JDBC & Databases-

1.SQL

- CRUD
- Joins
- Indexing
- Transactions

2.JDBC

- Connections
- PreparedStatement
- ResultSet
- Batch processing

3.Databases

- MySQL
- PostgreSQL

Design Patterns-

1.Principles

- SOLID
- DRY
- KISS

2.Creational

- Singleton
- Factory
- Builder

3.Structural

- Adapter
- Decorator
- Facade

4.Behavioral

- Observer
- Strategy
- Command

Advanced Java -

1. Reflection API
2. Annotations
3. Custom annotations
4. Generics deep dive
5. Networking
6. Sockets
7. JNI
8. Unsafe API basics